Then, $y_1^2 = 4ax_1$ and $y_2^2 = 4ax_2$,

and $y_1^2 - y_2^2 = 4a(x_1 - x_2)$.

Simplifying, $\frac{y_1 - y_2}{x_1 - x_2} = \frac{4a}{y_1 + y_2}$, the slope of the chord PQ.

Thus $(y - y_1) = \frac{4a}{y_1 + y_2}(x - x_1)$, represents the equation of the chord PQ.

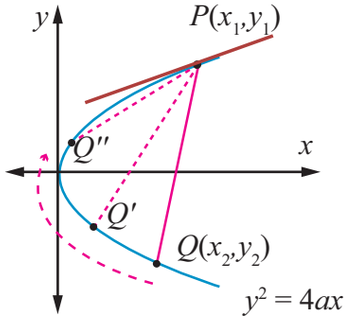


Fig. 5.48

When $Q \to P$, or $y_2 \to y_1$ the chord becomes tangent at P.

Thus the equation of tangent at (x_1, y_1) is

$$y - y_1 = \frac{4a}{2y_1}(x - x_1) \text{ where } \frac{2a}{y_1} \text{ is the slope of the tangent} \quad \ldots (1)$$

$$yy_1 - y_1^2 = 2ax - 2ax_1$$

$$yy_1 - 4ax_1 = 2ax - 2ax_1$$

$$\boxed{yy_1 = 2a(x + x_1)}$$

(ii) Equation of tangent in parametric form

Equation of tangent at $(at^2, 2at)$ on the parabola is

$$y(2at) = 2a(x + at^2)$$

$$\boxed{yt = x + at^2}$$

(iii) Equation of normal in cartesian form

From (1) the slope of normal is $-\frac{y_1}{2a}$

Therefore **equation of the normal** is

$$y - y_1 = -\frac{y_1}{2a}(x - x_1)$$

$$2ay - 2ay_1 = -y_1x + y_1x_1$$

$$xy_1 + 2ay = y_1(x_1 + 2a)$$

$$\boxed{xy_1 + 2ay = x_1y_1 + 2ay_1}$$

(iv) Equation of normal in parametric form

Equation of the normal at $(at^2, 2at)$ on the parabola is

$$x2at + 2ay = at^2(2at) + 2a(2at)$$

$$2a(xt + y) = 2a(at^3 + 2at)$$

$$\boxed{y + xt = at^3 + 2at}$$

Theorem 5.6

Three normals can be drawn to a parabola $y^2 = 4ax$ from a given point, one of which is always real.

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